

Unit 3 (Chapter 22.1) Review Answers

CHAPTER 22

✓ REVIEW

🔍 KEY QUESTIONS AND TERMS

1. a DOK 1
2. b DOK 1
3. The niche is the range of physical, temporal, spatial, and environmental conditions in which a population can survive and reproduce, as well as the method the species uses to obtain its resources and to reproduce. A habitat is analogous to a species' address, or an area that contains the physical and biological factors a species needs to survive. A niche is found within a habitat. DOK 2
4. Different species can live in the same tree by dividing the available resources in such a way that no species are competing with each other for the same resource. In this example, the birds could be eating different species of insects. DOK 1
5. When the population of predators increases, the population of prey decreases. After the population of predators decreases, the population of prey increases. DOK 2
6. The competitive exclusion principle states that no two species can share the exact same niche at the exact same time and place without one of them going extinct. This is because invariably one of the species outcompetes the other. DOK 1
7. In parasitism, one member of the relationship is harmed and one is benefited. In mutualism, both members benefit. In commensalism, one member benefits and the other is neither helped nor harmed. DOK 1
8. d, a, c, b DOK 2
9. d DOK 1
10. Primary ecological succession occurs in sites where there is no living organism present and very little organic material. Secondary succession occurs in a site where there are living remnants of the community present before the disturbance. DOK 1
11. The result of secondary succession is often a patchwork of communities in various stages of succession after disturbances that have occurred randomly with respect to time and location. (Students may also answer *climax community*.) DOK 1
12. It means first to settle. Pioneer species are the first to colonize new areas. They have broad ranges of tolerance for environmental factors. Some pioneer species break rock down into soil. DOK 2
13. c DOK 1
14. b DOK 1
15. Species diversity refers to the number of species in a designated area, whereas genetic diversity refers to the number of different genes (alleles) within a species. DOK 2

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22.1 Habitats, Niches, and Species Interactions

HS-LS2-2, HS-LS2-6, EP&CIIa

1. Which species directly affects the size and distribution of the plant population in its community?



a.



b.



c.



d.

2. Which of the following outcomes is most likely to occur if an organism's environment is outside the organism's range of tolerance?
 - a. The growth of the organisms increases.
 - b. The organisms fail to reproduce.
 - c. The organisms produce more offspring to increase the chance of survival.
 - d. The organisms develop mutualistic relationships.
3. What is the difference between an organism's habitat and its niche?
4. How can different species of insect-eating birds live in the same tree?
5. Describe how the population of predators changes as the population of prey changes.
6. What is the competitive exclusion principle?
7. What are the different types of symbiosis? For each, determine which organisms are benefiting, which are being harmed, and which are not affected.

1 Chapter 22 Communities and Ecosystem Dynamics

22.2 Succession

HS-LS2-2, HS-LS2-6, EP&CIIa, EP&CIIIc, EP&CIVc

8. Place these stages of succession in order from earliest to latest.



a.



b.

9. The first organisms to repopulate an area affected by a volcanic eruption are called
 - a. keystone species.
 - b. climax species.
 - c. primary producers.
 - d. pioneer species.
10. What determines whether ecological succession is classified as primary succession or secondary succession?
11. What is often the end result of secondary succession?
12. What is the meaning of the word *pioneer* in the term *pioneer species*? What are some common characteristics of pioneer species?

22.3 Biodiversity, Ecosystems, and Resilience

HS-LS2-2, HS-LS2-6, HS-LS2-7, HS-LS4-6, EP&CIIa, EP&CIIb, EP&CIIc, EP&CIVc

13. Which term describes the variety of species in a certain area?
 - a. succession
 - b. ecosystem
 - c. biodiversity
 - d. genetic diversity

16. An ecosystem service is a benefit that an ecosystem provides to humans. Some examples are: producing food, cycling nutrients, maintaining soil fertility, purifying water, storing carbon, pollinating crops, regulating pests, and buffering effects of extreme weather events. Students should explain how these services benefit humans. DOK 2
17. Ecosystem resilience refers to the ability of an ecosystem to maintain its structure and function in the face of a disturbance, such as the loss or decline of a species.

Planting a variety of crops can make a farm ecosystem more resilient to a change of conditions than if just a single crop species is present. DOK 2

18. The study of conservation biology focuses on protecting the biodiversity of life on Earth. It is an important field of study because all organisms, including humans, are part of the web of life. We need to preserve and protect all individuals and species on the planet. DOK 2